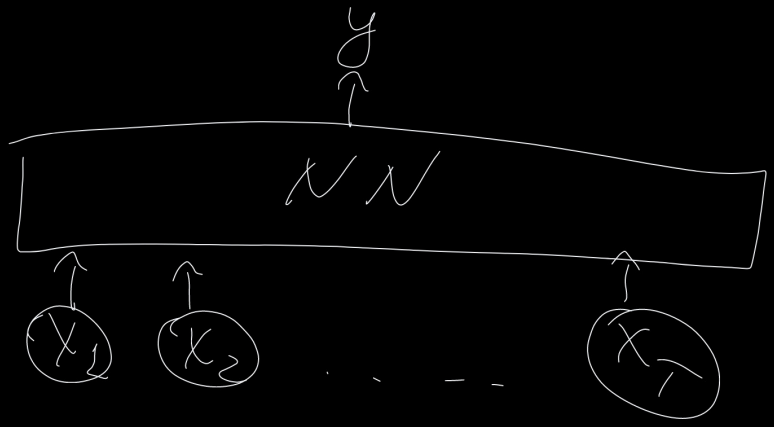
NNs for texts



Stemming

modelling
models
modelled

→ model

reg. ex.

Lemmatization

went → go

lem("modelling", pos="verb") = model

lem("modelling", pos="noun") = modelling

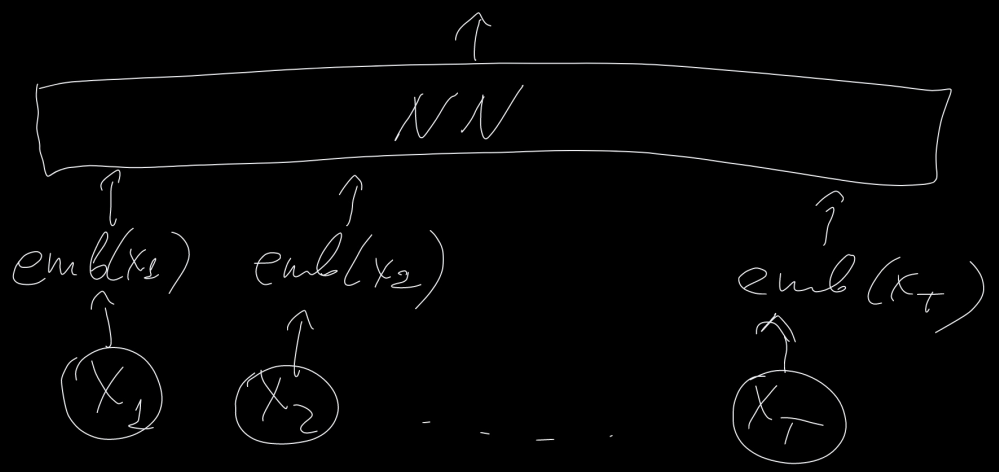
texts → preprocessing → words →

(to lower case
replace URLs with <URL>
remove commas)

→ stemming / lemmatization → stop-words removal → remove rare words

"the", "a", "in", "of" ...

Vocabulary: $x_t \in \{1, 2, \ldots, V\}$



$x_t \in \{1, \ldots, V\} \rightarrow$

$\rightarrow emb(x_t) \in \mathbb{R}^{emb_dim}$

BPE (Byte Pair Encoding) tokenizer

hug – 10
pug – 10
pun – 12
bun – 4
hugs – 5

No <UNK> token

Vocab: h, u, g, p, n, b, s

Step: hu – 15 Add
ug – 25
hg – 0

Step: hug – 15 Add
pu – 22

Inference:
hugs → h, u, g, s →
→ h, ug, s → hug, s

WordPiece

word → w, ##o, ##r, ##d

w, ##o → wo

##o, ##r → ##or

$$\text{score}(\text{token1}, \text{token2}) = \frac{\text{freq}(\text{token1 \& token})}{\text{freq}(\text{token1})\ \text{freq}(\text{token2})}$$

un, ##able

bre, ##men → bremen

Unigram tokenizer

$$p(w) = \sum_{z \in \text{TokenSplit}(w)} \prod_{t \in z} p(t)$$

hugs → h, u, g, s, hu, ug, gs, hug, ugs, hugs

$$\prod_i p(w_i) \to \max_{\{p(t)\}} \quad \text{– EM algorithm}$$

Step: throw away 10% of low-prob. tokens

Inference: $z \leftarrow \underset{z \in \text{TokenSplit}(w)}{\text{argmax}} \prod_{t \in z} p(t)$ ← Viterbi alg.

Sequence Piece

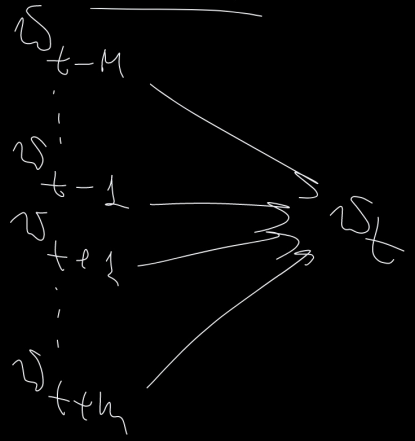
Hello world → _Hello_world → (BPE, Unigram) tokens

Word2Vec

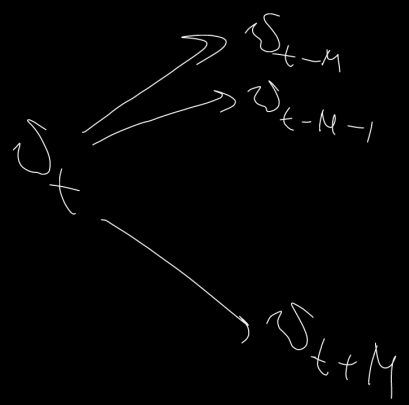
$text = [\ (w_1\ w_2\ w_3\ \ldots)\ \ldots\ \ldots\ \ldots\ ]$

sliding window

Ver. 1



Ver. 2



Вар. 1

$$Loss = \sum_t \frac{\exp\left(emb(x_t)^T \sum_{c \in C_t} emb(x_c)\right)}{\sum_{j=1}^{V} \exp\left(emb(x_j)^T \sum_{c \in C_t} emb(x_c)\right)} \to \max_{\{emb\}}$$

Вар. 2

$$Loss = \sum_t \sum_{c \in C_t} \frac{\exp\left(emb(x_t)^T emb(x_c)\right)}{\sum_{j=1}^{V} \exp\left(emb(x_j)^T emb(x_c)\right)} \to \max_{\{emb\}}$$

Negative sampling:

$$\sum_t \left[\sum_{c \in C_t} \sigma(emb(x_t)^T emb(x_c)) + \sum_{n \in N_t} \sigma(-emb(x_t)^T emb(x_n)) \right]$$

$\rightarrow \max_{\{emb\}}$

FastText

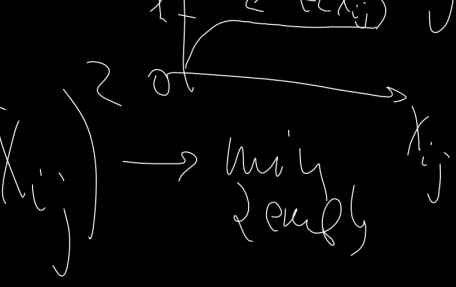
$$score(\omega_1, \omega_2) = \left(\sum_{h \in n\text{-grams}(\omega_1)} emb(h)\right)^T \left(\sum_{m \in n\text{-grams}(\omega_2)} emb(m)\right)$$

<where> → <wh, whe, her, ere, re>

GloVe

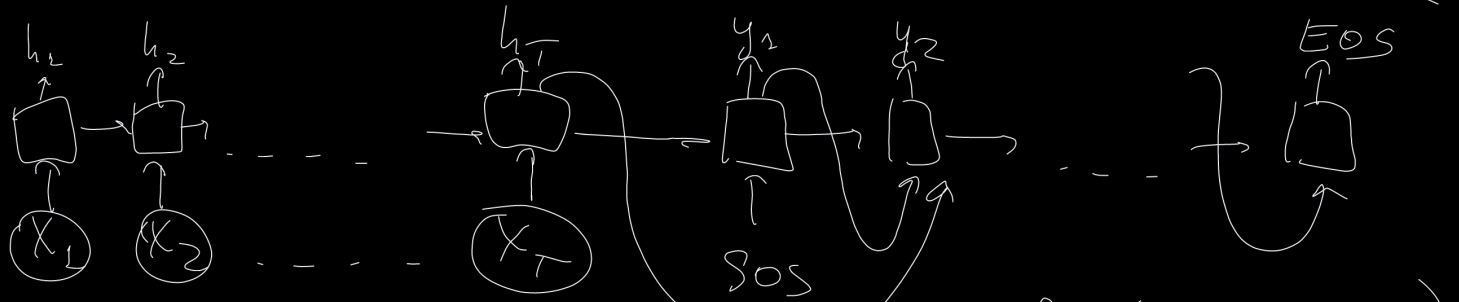
X_{ij} – # times token i occured together with token j

$$\sum_{i,j} f(X_{ij}) \left(emb(x_i)^T emb(x_j) + b_i + b_j - \log X_{ij}\right)^2 \to \min_{\text{emb}}$$



CoVe

machine translation task

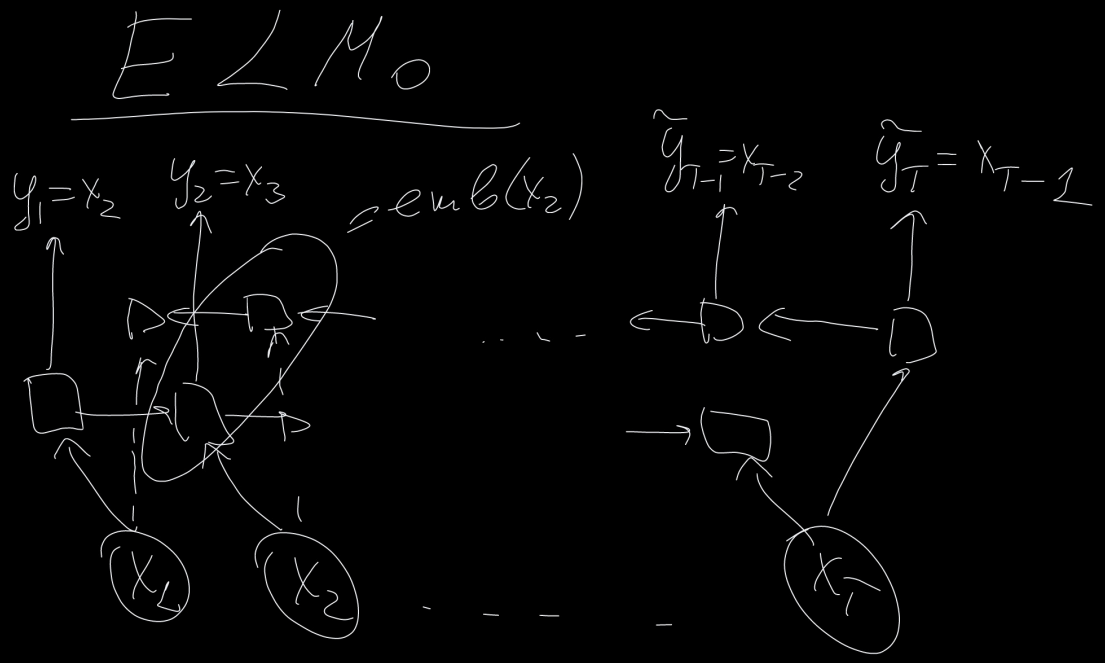


Teacher Forcing: input $y_{true,t}$ to decoder

Exposure bias: with p give $y_{true,t}$ as input

1-p give y_t

ELMo



BERT-like embedding

Beam search

$$p(y \mid x, \theta) = p(y_1 \mid x, \theta)\, p(y_2 \mid y_1, x, \theta) \cdots p(y_n \mid y_1 \cdots y_{n-1}, x, \theta)$$

Inference: $y = \arg\max_y p(y \mid x, \theta)$

$K = 2$

$t=1$	$t=2$	$t=3$
top-K(a −0.5	top-K(aa −0.5·0.5	aaa
b −0.4	ab −0.5·0.4	aab
c −0.1	ac −0.5·0.1	aac
	ba −0.4·0.33	aba
	bb −0.4·0.33	abb
	bc −0.4·0.33	abc

$$\frac{\log p(y \mid x, \theta)}{\text{length}(y)^{\alpha}}$$

$\alpha \in (0, 1]$

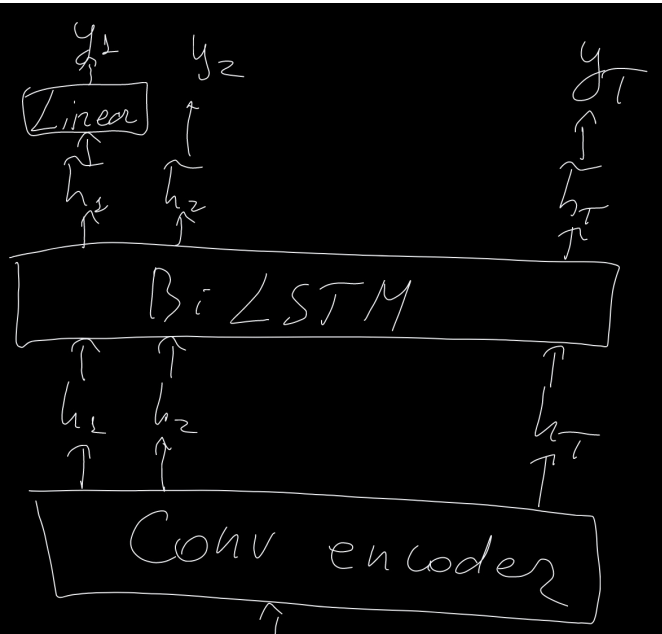
CTC-loss

handwritten text recognition



image

$\rightarrow h, e, l, l, o$

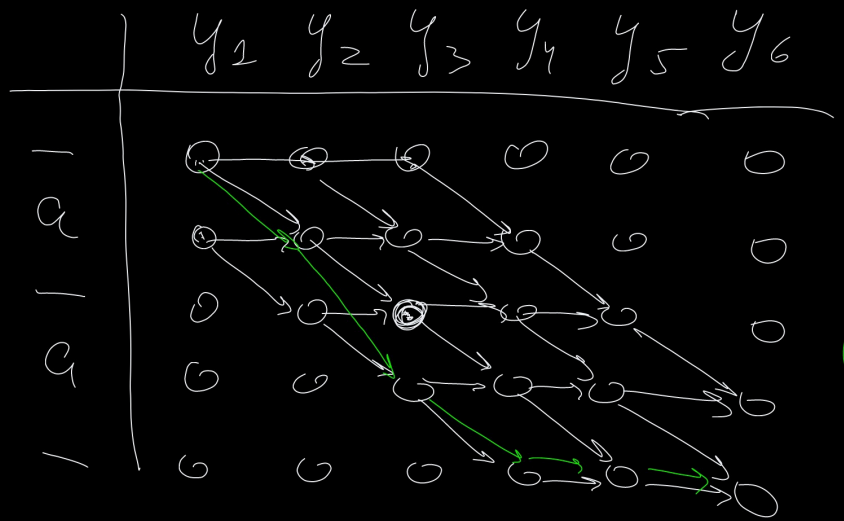


$$P_{CTC}(y_{true} \mid y_1 \dots y_M, X, \theta) = \sum_{y:\, CTC(y) = y_{true}} \prod_{t=1}^{T} P_t(y_t \mid X, \theta)$$

$$CTC(__h_ee_l_l_oo) = _h_e_l_l_o \rightarrow$$

$\rightarrow$ hello

$y_{true} = "aa"$, $T = 6$



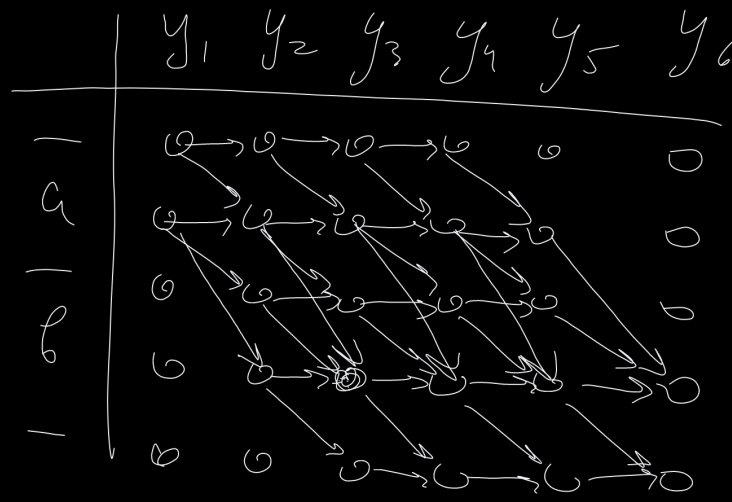
$CTC(___a_a) = aa$

$CTC(_aa___) = a$

$\alpha_{s,t} = p_t(y_t = y_{true,s} | X, \theta)(\alpha_{s,t-1} + \alpha_{s-1,t-1})$

Comp. complexity: $O(length(y_{true}) \cdot T)$ $CTC = \alpha_{4,6} + \alpha_{5,6}$

$y_{true} = "ab", \quad T = 6$



$$\alpha_{s,t} = p_t(y_t = y_{true,s} | x, \theta) \times \left(\alpha_{s,t-1} + \alpha_{s-1,t-1} + \alpha_{s-2,t-2} \right),$$

if $y_t \neq _$ and $y_{true,s} \neq y_{true,s-2}$

$$\alpha_{s,t} = p_t(y_t = y_{true,s} | x, \theta) \left(\alpha_{s-1,t} + \alpha_{s-2,t-1} \right)$$

otherwise

Non-differentiable loss functions

$BLEU(y_1, y_2)$ – text translation

$CIDEr(y_1, y_2)$ – image captioning

$$F(\theta) = \mathbb{E}_{p(y|x,\theta)} \overleftarrow{f(y)} \to \max_{\theta}$$

← non-diff.

REINFORCE / Log-derivative trick

$$\nabla_\theta F(\theta) = \nabla_\theta \mathbb{E}_{p(y|x,\theta)} f(y) =$$

$$= \nabla_\theta \sum_y p(y|x,\theta) f(y) = \sum_y \nabla_\theta p(y|x,\theta) f(y) =$$

$$= \sum_y p(y|x,\theta) \underbrace{\frac{\nabla_\theta p(y|x,\theta)}{p(y|x,\theta)}}_{\nabla_\theta \log p(y|x,\theta)} f(y) = \mathbb{E}_{p(y|x,\theta)} \nabla_\theta \log p(y|x,\theta) f(y) \approx$$

$$\approx \frac{1}{M} \sum_{j=1}^{M} \nabla_\theta \log p(y_j|x,\theta) f(y_j), \quad y_j \overset{i.i.d.}{\sim} p(y|x,\theta)$$

Adding baseline $b(x)$

$$\nabla_\theta F(\theta) = \mathbb{E}_{p(y|x,\theta)} \nabla_\theta \log p(y|x,\theta)\big(f(y) - b(x)\big)$$

$$\mathbb{E}_{p(y|x,\theta)} \nabla_\theta \log p(y|x,\theta)\, b(x) = b(x) \sum_y p(y|x,\theta) \nabla \log p(y|x,\theta) =$$

$$= b(x) \sum_y \cancel{p(y|x,\theta)} \frac{\nabla_\theta p(y|x,\theta)}{\cancel{p(y|x,\theta)}} = b(x) \nabla_\theta \underbrace{\sum_y p(y|x,\theta)}_{1} = 0$$

Good baseline: ① $b(x) = \mathbb{E}_{p(y|x,\theta)} f(y)$

② $b(x) = f(y_{\text{greedy rollout}})$

$$L(\theta) := -\frac{1}{M} \sum_{j=1}^{M} \log p(y_j|x,\theta)\,(f(y_j) - b(x)) \to \min_\theta$$

$$\theta = \text{opt_step}(\nabla_\theta L) \qquad y_j \sim p(y|x,\theta)$$